

Dhaneswara Mandrasa

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PROFESSIONAL SUMMARY

Senior Data Scientist & AI Engineer with 8+ years across telecom, banking, commerce and geospatial: ML, causal inference and LLMs for pricing, risk and mobility. Three years in Abu Dhabi at Eureka AI working with 7 telecom operators in 7 countries: segmented 100M+ daily users for clients including Gojek, Grab and Tokopedia, built a credit score with the company's best-ever Gini (0.6) and grew the products I led from 5% to 70% of revenue. Shipped a production LLM system on Snowflake Cortex and led data & AI for a 30K-merchant commerce platform. Now building LLM feature-extraction pipelines for causal and demand models at Hiroshima University.

RESEARCH

Master's researcher in Smart Mobility, Hiroshima University. I build LLM feature-extraction pipelines that turn unstructured geospatial text into reliable, validated features for causal and demand models. Current work: using NYC's January 2025 congestion toll as a natural experiment on millions of ride-hail trips, I test whether LLM-scored place features improve price-response and welfare estimates against a no-LLM baseline.

LLM features for causal pricing evaluation · Urban and Data Science Lab, Hiroshima University **2026 – present**

- **LLM feature extraction / context engineering:** converted OpenStreetMap POIs and place names into structured prompts, and used Llama-3 70B to score 15 semantic features across 4 construct families (activity, physical environment, safety, built environment) for 262 NYC zones.
- **LLM evaluation and reliability harness:** built a versioned prompt library with staged wording probes, repeated-run scoring (ICC = 0.930), a frozen, hashed prompt spec, and a scorer chosen by training cutoff so it could not have seen the policy period, which rules out leakage.
- **Causal impact / geo-experiment:** ran a difference-in-differences (incrementality) analysis of the congestion toll on millions of NYC TLC high-volume trip records and estimated a -2.6 to -2.9 pp shift in ride-hail share in tolled zones.
- **Ablation:** do LLM features add value? Designed a 4-arm ablation ladder (no-LLM baseline → flat LLM features → latent-factor model → embeddings) that measures incremental lift, with a SHAP-explained gradient-boosting check.
- **Demand and price elasticity modelling:** built a discrete-choice demand model (hybrid choice / ICLV) whose price coefficient gives price elasticity and willingness-to-pay by zone segment.
- **Data engineering:** built a bronze/silver/gold pipeline over millions of trip records.

PROFESSIONAL EXPERIENCE

Olsera – cloud POS & commerce platform, 30K+ merchants across Indonesia

Jakarta, Indonesia

Head of Data & AI

Aug. 2025 – Mar. 2026

- Owned the data & AI roadmap for a platform processing hundreds of thousands of transactions a day, reporting to executives; hired and led the data team.
- Built the data function from scratch: architecture and schemas for orders, products, roles and logs behind analytics and AI use cases.
- Led the platform decision with proof-of-concept evaluations of Snowflake vs Databricks, benchmarked against BigQuery and Redshift; prototyped AI stock recommendations for merchants.

iZeno – enterprise IT & data consulting

Jakarta, Indonesia

Lead Data Scientist

Jul. 2024 – Mar. 2025

- **Cut the run time of a credit-scoring data and ML pipeline by 75%.**
- **Built a production LLM news-analytics system on Snowflake Cortex for a financial-sector client, cutting misclassification by 27% with a knowledge graph of 10K+ entities.**
- **Designed a hybrid recommendation engine for a bank that improved targeting accuracy by 35%, and worked with domain experts on a root-cause analysis system (+18% yield).**

Eureka AI – telecom data & AI, 7 operators in 7 countries

Abu Dhabi, UAE

Senior Data Scientist

Apr. 2021 – Jul. 2024

- **Scale:** worked with 7 telecom operators in 7 countries (Indonesia, Malaysia, Singapore, Thailand, Vietnam, Hong Kong, UAE) on hundreds of millions of rows, in a multinational team from Singapore, Indonesia, Malaysia, India, the UAE and the US.
- **Segmentation:** segmented 100M+ daily active ride-hailing and e-commerce users into hierarchical segments with supervised and unsupervised models (k-means, Random Forest, XGBoost) on clickstream data in PySpark ML and MLflow; models served 5 major digital-platform clients including Gojek, Grab, Tokopedia and foodpanda.
- **Personas:** identified 30+ personas for 80M+ monthly active telecom subscribers from clickstream, mobility and subscriber data, with a Word2Vec look-alike model in Spark; both became reusable products across clients.

- Credit risk: built an alternative credit score from telco and clickstream data reaching a Gini of 0.6, the highest in the company's history.
- Product & revenue: grew the Market Intelligence products from 5% to 70% of company revenue in under two years; presented models to the Chief Product and Chief Commercial Officers and supported the commercial team as technical lead in client pitches.
- Team: hired, mentored and set modelling standards for a data science team of 6; made models explainable to business users with SHAP.

Bhumi Varta Technology – location intelligence

Jakarta, Indonesia

Data Scientist

Jun. 2020 – Apr. 2021

- Built spatial models on POI, mobile and sales data that guided clients' expansion from under 100 to 200+ outlets across Indonesia.
- Estimated poverty levels from satellite imagery with CNNs (PyTorch), refining socio-economic status resolution from 1×1 km to 100×100 m.

AEON Indonesia – retail

Jakarta, Indonesia

Management Trainee

Apr. 2018 – Sep. 2019

- Ran sales and operations analysis during a cross-functional rotation (store operations, merchandising, marketing, supply chain), turning store data into process-improvement recommendations.

SKK Migas – upstream oil & gas regulator

Jakarta, Indonesia

Data Analyst

Apr. 2016 – Apr. 2018

- Built an early ML model predicting oil & gas resources for 200+ working areas (<10% test error); migrated data from 50+ companies; built Python/Tableau dashboards.

EDUCATION

Hiroshima University

Hiroshima, Japan

MPhil, Smart Mobility (AI and data science for transport)

Apr. 2026 – Mar. 2028 (expected)

Urban and Data Science Lab, supervised by Prof. Tao Feng. Programme: Innovation and Practice for Smart Society.

Institut Teknologi Bandung (ITB)

Bandung, Indonesia

B.Sc. Geological Engineering – GPA 3.52/4.00, Cum Laude

Aug. 2011 – Aug. 2015

TECHNICAL SKILLS

Programming: Python (pandas, polars, NumPy, scikit-learn, statsmodels), SQL, R, PySpark

AI & LLM stack: LangChain, LangGraph, Snowflake Cortex, Llama-3, RAG (Elasticsearch), prompt design and evaluation, knowledge graphs, FastAPI, spaCy, PyTorch

Machine learning: supervised and unsupervised learning, XGBoost and gradient boosting, Random Forest, clustering and segmentation, recommendation systems, credit scoring, CNNs, SHAP, MLflow

Data platforms: Snowflake, Databricks, Apache Spark / Spark ML, Dataiku, Airflow, medallion (bronze/silver/gold) pipelines

Cloud & DevOps: AWS, GCP, Azure, Docker, Git

Causal inference & econometrics: difference-in-differences, discrete choice (MNL, NL, ICLV) in Biogeme, factor analysis, price elasticity and welfare estimation

Geospatial: QGIS, GeoPandas, OpenStreetMap, satellite imagery, POI and mobility data

LANGUAGES

English (advanced); Bahasa Indonesia (native)